AI at UFlex: more from every kilogram while new capacity comes on stream

UFlex is in a growth phase. Egypt aseptic is in commercial trials, a 54,000-tonne BOPP line is coming at Dharwad, and Mexico WPP and the Noida recycling plant came on stream in 2026. Film lines ran at 85.4% utilisation in Q1 FY27. This note sets out where AI can add value in that phase, and a way to test it on one film line before any larger spend.

₹18 cr

Stage 1 commitment, before any roll-out decision

range ₹16 cr to ₹22 cr

1.02%

of film output removed as scrap on the pilot line: where the film-line roll-out pays for itself

with 3.4% energy saved per good kg

₹34 cr

recurring benefit in year 5 (base); ₹14 cr after the programme's own running cost

plus one-off: ₹61 cr inventory cash, ₹8 cr ramp-up

WHERE AI HELPS

Six use cases, valued on tonnes rather than revenue

USE CASE	WHAT AI ADDS	BASE VALUE	WHEN
1 · Film-line yield	Links breaks, gels and specks to upstream causes, from chip IV and moisture to the winder; warns shift leads before a break; tracks energy per good kg	₹17.9 cr a year	STAGE 1 PILOT
3 · Recycling quality	Grades bales from images and lab data; predicts rPET yield and grade; keeps records from bale to pack that fit a recognised chain-of-custody model	₹15.8 cr a year*	STAGE 1
4 · New-line ramp-up	Captures start-up and grade-qualification data at Dharwad from commissioning, rather than as a retrofit	₹8.3 cr one-off	STAGE 1
5 · Inventory planning	Forecasts by grade, market and customer; inventory norms by item and plant; price-risk signals for PTA, MEG and polypropylene	₹61 cr cash, one-off	STAGE 1
2 · Slit and schedule	Grade and thickness campaigning, changeover-loss prediction and BOPP master-roll ageing, on top of existing deckle optimisation	₹3.3 cr a year	AFTER GATE 2
6 · Print make-ready	Press and ink settings from job history and spectral colour data; viscosity drift; returned-ink reuse	₹4.3 cr a year	AFTER GATE 2

Base case, pre-tax, at full rate. *At full recycling utilisation; ₹11.9 cr at 75%. Inventory value counts half of a five-day release as attributable to AI; interest on released cash is not counted. Maintenance, metallising, chip plants and aseptic laminate integrity are sized in the sprint, not here.

WHY NOW

Recycled-content records are becoming a customer requirement

India's Plastic Waste Management (Amendment) Rules 2026 raise recycled-content obligations for flexible packaging from 10% to 20%, and for multilayer from 5% to 10%, by 2027-28. QR labelling has applied since July 2025, and the EU's Packaging and Packaging Waste Regulation has applied since August 2026. Brand owners will ask suppliers for records by plant and format. UFlex makes chips, film, inks and packs and recycles within one group, so it can assemble those records from resin to pack in one place.

HOW THE MONEY IS STAGED

Commit ₹18 cr first; the larger spend waits for measured results

Stage 1 covers one pilot film line chosen by UFlex, inventory planning, recycling quality at Noida and Dharwad instrumentation. None of the enterprise use cases depends on the pilot line.

The film-line roll-out follows only if the pilot line, measured for six months against a control line, clears its own break-even on its own costs. In the base case that is about 1.02% of film output removed as scrap and 3.4% energy saved per good kilogram, 1.7 times our base rates.

At conservative rates the full programme does not pay back within five years on pre-tax cash. It is positive in the base case once steady state after year 5 or the inventory release is included. That is the reason for staging it.

₹ CRORE	LOW	BASE	HIGH
Stage 1 commitment	22	18	16
Year-5 recurring benefit	15	34	64
Year 5, net of running cost	−11	14	45
5-year NPV, pre-tax cash	−96	−30	50
NPV with steady state after year 5	−136	35	331
NPV with inventory release	−106	86	415

Costs include 25% contingency, plant time, licences, sensors and FX. The low case also assumes a 25% overrun. Build cost is charged when spent.

GATES

Each commitment follows a measured result, signed by group finance

Gate 1 · week 2	A line and a problem chosen, the data request agreed, and the AI, inspection and line-supplier digital services already installed listed, so nothing is bought twice.
Gate 2 · month 10	The pilot line against its control, normalised for grade mix, reaches its break-even. Below it, UFlex keeps the enterprise use cases and the data foundation and does not roll out the film-line work.
Gate 3 · month 18	India sites tracking the measured rates in the value ledger before any commitment to overseas sites.

ARCHITECTURE

One data foundation, built once, on what UFlex already runs

Plant and ERP
Line controls, OEM, lab, energy, ERP

→

Plant edge
Read-only exports by UFlex IT

→

Data foundation
Site, line, grade, order, roll; resin to pack

→

AI services
Cause-finding, break warning, forecasts

→

Applications
Shift-lead advice, value ledger

Security across all layers: no remote access to plant networks; IEC 62443 segmentation before any integration

NEXT STEP

A two-week readiness sprint on one film line and one problem UFlex chooses, preceded by a short working session with film technical leadership. Fixed fee; UFlex owns the deliverables, with no exclusivity and no obligation to continue.

The sprint, day by day

Days 1–3 immerse on the nominated line and interview leaders. Days 4–7 audit data and the IT and OT landscape. Days 8–11 rebuild the model on UFlex numbers and agree the Gate 2 test with group finance. Days 12–14 decide the pilot scope together.

About PLUMB Advisory

Independent AI strategy: no reseller, referral or implementation fees from any technology vendor. Pilot fees are capped, with part held back until Gate 2 is signed off. Film-process expertise: ***[to confirm: named specialist]***.

Prepared from public information only: UFlex annual reports FY25–FY26, the Q1 FY27 investor presentation and industry sources. Figures will be refreshed after the Q2 FY27 results. A full strategy with sources and the business-case workbook is available on request.

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